Replicating Braboszcz et al. (2017) on OpenNeuro `ds003969`: pipeline decisions, sensitivity, and a look at criticality

October 2026

Abstract

I rebuilt the EEG pipeline of Braboszcz et al. (2017, *PLoS ONE* 12:e0170647) in MNE-Python and applied it to the public release of their data, restricted to the paper's own age- and gender-matched sample, which I recovered from the authors' Zenodo deposit. With no ICA component removal, the replication reproduces the paper closely: per-subject parieto-occipital gamma correlates with the paper's published values at $r = 0.97$, and the group differences agree to within 0.6 dB. Both headline results replicate. The gamma (60–110 Hz) excess in meditators over controls is about 4 dB during meditation. Vipassana practitioners have more 7–11 Hz alpha than every other group. Automated ICA component removal shrinks the gamma effect, from about 4 to 2.5 dB, though the pooled meditator–control difference stays significant. The paper's own values track the no-ICA data, so its manual ICA left parieto-occipital gamma largely untouched. Whether the extra power removed by automated ICA is artefact or signal remains open. I then computed long-range temporal correlations (DFA) and functional E/I (fEI) with the parameters of the recent preprint "Near-critical brain dynamics track effortlessness during meditation". As in that preprint, there are no meditator–control differences. The one apparent within-meditator effect (beta DFA higher in the second meditation block than in the first) appears equally in controls. It is therefore best read as a time-on-task effect, not as an effect of meditative depth.

1 Data and target results

The dataset (`ds003969`) contains 64-channel Biosemi recordings of 98 participants: 31 controls (CTR), 24 Himalayan Yoga (HTR, "HYT" in the paper), 20 Isha Shoonya (SNY, "ISY"), 19 Vipassana (VIP) and 4 Transcendental Meditation (TM). Each participant did a 20-minute instructed mind-wandering (IMW) session and a 20-minute meditation session (MED). These are released as four ∼10-minute files: `think1`, `think2`, `med1breath` (breath focus) and `med2` (the tradition's own practice; controls keep doing breath focus). The paper analysed the last 10 minutes of each session, which here are `think2` and `med2`.

The recording start times in the BDF headers show how the blocks were scheduled. Both blocks of a session were recorded back to back (median gap under a minute), and the two sessions were separated by a median of 66 minutes (range 47–89), all on one day. Half of the participants did the MED session first and half the IMW session first, as recorded in the `first_session` column of `participants.tsv`; this agrees with the timestamps for all 98. Within a session the order is fixed, so `med2` always directly follows `med1breath`. (The dataset README describes the blocks as alternating, which the timestamps contradict.)

The paper analysed 64 participants, 16 per group, matched on age and gender from a larger pool. The dataset does not record which participants these were. The authors' earlier Zenodo deposit (doi 10.5281/zenodo.57911) lists them by lab ID, along with their per-subject median gamma and alpha values. I mapped these lab IDs to `ds003969` subjects using the recording dates in the BDF headers, cross-checked against age. 63 of the 64 are in `ds003969`; HYT26's recording was never converted. All results in this report use this matched sample (16 CTR, 15 HYT, 16 ISY, 16 VIP). The pipeline restricts to it with `run_pipeline.py --paper-subjects`.

The results targeted were:

1. **Gamma (60–110 Hz).** Higher parieto-occipital power in all three meditator groups than in controls, both as a trait (MED and IMW combined) and during MED (the paper's Fig. 3, reproduced as Fig. 4).
2. **Alpha (7–11 Hz).** During MED, VIP higher than CTR, ISY and HYT.
3. (Not re-analysed; see the note in Section 4) the correlation between gamma and log practice hours (Fig. 4D).

2 Pipeline: decisions and likely discrepancies

Table 1 sets the paper's description against what was implemented. Every step was either matched as closely as the text allows or replaced by an automated equivalent, because the original used manual judgement. Counts in the table are for the 252 matched-sample recordings.

Step	Paper	This replication
Sample	16 per group, matched on age and gender from a larger pool; TM not included	The paper's matched list, 63 of 64 available (Section 1)
Reference	Right mastoid	Right mastoid (`EXG6`). `Fp1` replaced by `EXG8`, the spare electrode wired in its place
Resample, filter	256 Hz; 1 Hz IIR high-pass, order 6, 0.3 Hz transition	Same
Segment rejection	EEGLAB `pop_rejcont`: 1 s epochs, 0.5 s overlap, 0–10 Hz >17 dB, 35–128 Hz >14 dB, contiguous 8; thresholds "in rare cases" adjusted by 1–5 dB after visual review	Line-by-line Python port with unit tests, including EEGLAB quirks such as `contiguous` acting as a merge distance. Manual threshold adjustment **not reproduced**
Bad channels	Manual; 0–18 per subject, mean ≈5; interpolation not stated	Automated: union of MNE waveform LOF (thr. 1.5) and a custom spectral LOF on 1–45 Hz log-PSD (thr. 2). Mean 6.8 per recording, median 6, range 0–30. Not interpolated; bad channels are excluded and cluster means use the remaining channels
ICA	Infomax; 1–5 eye/temporal-muscle components chosen visually by one experimenter	Infomax with 20 components (count not given in the paper), fit on good EEG channels. Components chosen automatically with `find_bads_muscle` (thr. 0.75) and `find_bads_eog` (z>3), capped at ≥1 muscle, ≤4 EOG and ≤5 in total: mean 3.3 removed (2.4 muscle, 0.9 EOG), range 1–5. A second pipeline with no ICA is reported alongside (Section 3)
Spectra	1 s non-overlapping epochs, Hanning taper, FFT, $10\log_{10}$; 2–110 Hz, 50 Hz band excluded	Same epoching and taper (Welch with one segment per epoch); epochs overlapping `BAD` spans dropped; linear power averaged over epochs, then dB
Statistics	FieldTrip cluster permutation on log power (2000 resamples) over electrodes; Welch/mixed ANOVA	Welch t-tests on the mean over an 8-electrode parieto-occipital cluster (O1, Oz, O2, PO7, PO3, POz, PO4, PO8); no cluster correction

Table 1: Paper pipeline compared with the replication.

Where results are most likely to diverge. In rough order of expected impact:

1. **ICA component selection.** The paper removed 1–5 components chosen by eye. The automated selection controls the size of the gamma result (Section 4).
2. **Bad channels.** The automated detector removes slightly more channels than the paper did (6.8 vs ≈5 on average). Channels are dropped, not interpolated, so the parieto-occipital average is taken over a cluster whose make-up varies from subject to subject.
3. **Statistics.** The cluster permutation approach was not re-implemented. The tests here are uncorrected tests on one predefined cluster, so their p-values are not directly comparable to the paper's.
4. **Unreproducible manual steps.** The paper's `pop_rejcont` threshold tweaks and its channel and component choices cannot be recovered from the text.

Two smaller points came up along the way:

- The paper's y-axis (∼45 dB at low frequencies) is offset from a correctly scaled PSD in μV^2/Hz by a constant of roughly 35–45 dB. This is consistent with an unnormalised squared FFT magnitude. Because it is a constant offset in dB, between-group differences are unaffected.
- The 100 Hz line-noise harmonic falls inside the gamma band and is not excluded. Visual inspection showed nothing abnormal at 100 Hz.

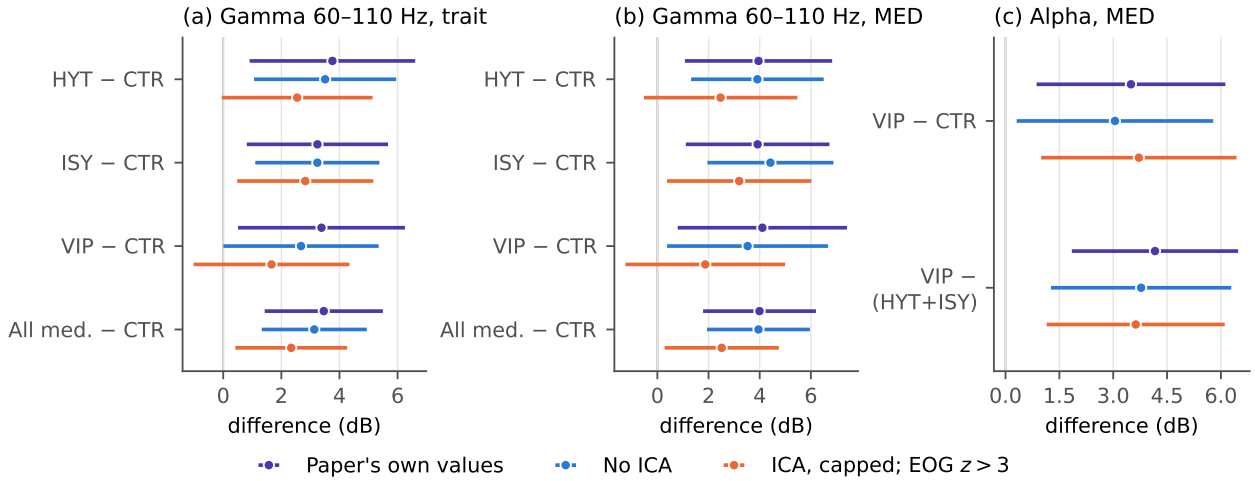


Figure 1: Group differences (dB) with 95% CIs from Welch t-tests, for this replication's two pipelines and for the paper's own per-subject values (its "Posterior" measure; trait = mean of MED and MW).

3 How the final pipeline was reached

Two decisions changed during the project. Results from the superseded versions are not reported here.

Sample. The project was first run on all 94 non-TM participants, because the dataset does not say which participants the paper analysed. That sample is not the paper's: it has about 50% more participants, is not age-matched, and its additional controls are younger than the paper's. Its group comparisons therefore do not test the paper's claims, and they gave a misleading picture of how well the paper replicates. Recovering the paper's subject list from its Zenodo deposit (Section 1) fixed this, and the spectral and criticality pipelines were rerun on that list. Because each recording is processed independently, the matched-sample spectra are identical (0.000 dB difference) to the earlier runs' values for the same subjects; only the participants compared have changed.

ICA. The paper's component choices were made by eye and cannot be recovered, so several automated rules were tried while developing the pipeline:

- An uncapped rule (muscle threshold 0.5, EOG z>3) often removed more than 10 of 20 components per recording, far more than the paper's 1–5.
- Raising the thresholds (muscle 1.0, EOG z>5) removed essentially nothing.
- An intermediate uncapped rule (muscle 0.75, EOG z>3) removed a median of about 2 but up to 17.
- The rule adopted caps the count to mirror the paper: at least one muscle component, at most four EOG and at most five in total. Variants with a higher EOG threshold (z>4) and with the EOG channels filtered before scoring were also tried.

Because the gamma result proved sensitive to ICA, the final analysis runs the capped rule (`_matchsample_v0`) and a no-ICA pipeline (`_matchsample_noica_v0`) side by side. A manual labelling of components was started on 19 recordings (`processed_manual`), and the pipeline now has a manual mode for this.

4 Spectral results

The paper's own per-subject values from Zenodo were put through the same tests as the replication's two pipelines. This separates differences in processing from differences in statistics.

Without ICA, the paper is reproduced. Table 2 and Fig. 1 show the no-ICA run agrees with the paper's own numbers to within 0.6 dB for every gamma contrast. All three meditator groups exceed controls during MED, and the four-group ANOVA is significant. The match also holds subject by subject (Fig. 2a):

- Gamma: $r = 0.97$ in MED and 0.93 in MW.
- Alpha: $r = 0.97$.
- The residual scatter around a constant offset is 1.1–1.7 dB per subject.

		Capped ICA	No ICA	Paper's values
Gamma, MED	HYT − CTR	+2.5 (.12)	+3.9 (.007)	+4.0 (.012)
	ISY − CTR	+3.2 (.034)	+4.4 (.001)	+3.9 (.011)
	VIP − CTR	+1.9 (.25)	+3.5 (.038)	+4.1 (.023)
	All meditators − CTR	+2.5 (.034)	+4.0 (<.001)	+4.0 (.001)
	Welch ANOVA, 4 groups	$F = 1.8$ (.16)	$F = 5.2$ (.005)	$F = 4.0$ (.016)
Gamma, trait	All meditators − CTR	+2.3 (.023)	+3.1 (.002)	+3.5 (.002)
Alpha, MED	VIP − CTR	+3.7 (.012)	+3.1 (.037)	+3.5 (.014)
	VIP − (HYT+ISY)	+3.6 (.007)	+3.8 (.006)	+4.2 (.001)

Table 2: Differences in parieto-occipital power (dB) with Welch p-values, uncorrected. Source: `results/matched/matched_group_tests.csv`.

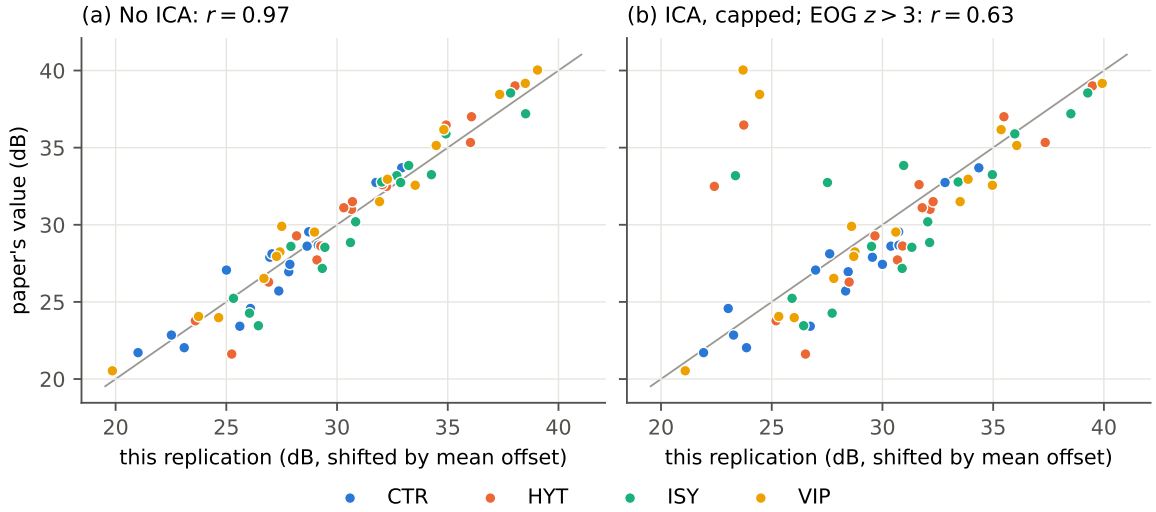


Figure 2: Per-subject parieto-occipital gamma during MED: this replication vs the paper's published values, with the replication shifted by its mean offset of 157 dB. Of this, 120 dB converts V^2/Hz to $\mu V^2/Hz$ and the rest is the FFT scaling noted in Section 2. The line is identity.

This is strong evidence that the reimplemented pipeline is faithful to the original, from referencing through segment rejection to spectral estimation. It is a reproduction on the same data, not an independent replication.

Automated ICA attenuates gamma. With the capped ICA, the gamma contrasts shrink:

- The pooled meditator–control difference falls from about 4 to 2.5 dB and remains significant ($p = .034$ in MED, $p = .023$ trait).
- The four-group ANOVA is no longer significant.
- Only ISY survives individually.

The attenuation is concentrated in a few recordings. In MED the median subject loses only 0.2 dB, but 8 of 63 subjects lose more than 3 dB (3.9–16.9 dB), and 7 of those 8 are meditators (Fig. 2b). These recordings did not have unusually many components removed: half of them lost only one or two, a mix of muscle and EOG components. Across all recordings the number of muscle components removed is similar between groups (1.9–2.6 per recording). So the attenuation comes from *what* a few components contain in a few meditators, not from removing more components in meditators.

The paper's manual ICA evidently did not remove this power, because its values track the no-ICA data. Two readings fit:

1. The automated detectors correctly remove EMG or eye-related activity that is present in these meditators. The paper's gamma effect would then be partly artefactual.
2. The detectors also capture genuine broadband high-frequency activity, or microsaccadic spike potentials, which a spectral-slope muscle criterion cannot tell apart from EMG.

The paper's own control analysis is relevant here: its eye-component cluster correlated with parieto-occipital gamma ($r = 0.33$), whereas temporal-muscle components did not. Labelling the components removed in these 8 recordings by hand is the direct test.

Alpha replicates. VIP exceeds CTR by 3.1–3.7 dB ($p = .012$–$.037$), matching the paper's +3.5 dB. VIP also exceeds the other two traditions by 3.6–3.8 dB ($p \leq .007$), under both pipelines.

Age. Within the matched sample (no ICA, trait gamma), gamma tends to *fall* with age in controls (Spearman $\rho = -0.45$, $p = .08$, $n = 16$) but *rise* with age in meditators ($\rho = +0.26$, $p = .08$, $n = 47$). Neither trend is significant on its own, but together they mean the size of the group difference depends on the age range sampled. Age-matching, as the paper did, is therefore essential to the comparison.

A note on the experience correlation (Fig. 4D). Practice hours rise with age, so the age trend in meditators bears on the paper's correlation between gamma and log practice hours. In the paper's own per-subject data for the 48 meditators:

- Gamma correlates with age ($r = 0.27$, $p = .07$) more than with log hours ($r = 0.16$, $p = .27$).
- Log hours and age correlate with each other ($r = 0.38$).
- With age in the model, log hours explains nothing ($p = .63$).

The published correlation used a skipped correlation over a different subset of conditions, so this is not a re-analysis of it. It does suggest the experience effect cannot be separated from age in this sample.

5 Criticality: DFA and fEI

Method. The Hilbert amplitude envelope was computed in theta, alpha (8–13 Hz), beta (13–30 Hz), broadband (1–30 Hz) and gamma (60–110 Hz). DFA and fEI used `crosci` on non-overlapping 180 s epochs, with DFA fitted over 5–30 s and fEI computed in 5 s windows with 80% overlap. Values were averaged over epochs and good channels to give one value per subject, task and band. Manual implementations of DFA and fEI were first written and checked against `crosci`; they differed slightly because of windowing.

Contrasts were fitted as linear mixed models with a random intercept per subject, for both pipelines (`scripts/matched_criticality.py`; results in `results/matched/matched_criticality.csv`).

Data loss. An epoch is dropped if *any* `BAD` span overlaps it. Even a 2 s `pop_rejcont` flag therefore discards 180 s of data. Consequences:

- Only 149 of 252 recordings (59%) have at least one usable epoch, from 55 of the 63 subjects.
- `think2` retains only 28 of 63 recordings.
- Usable recordings average 1.7 epochs, about 5 of 10 minutes.
- Only 26 meditators and 6 controls have both meditation blocks. The mixed models also use subjects with one block, but the within-subject contrasts rest mainly on these.

This is the most important practical weakness of the criticality analysis. An epoch-length sweep on the same recordings (`exploration/261012_*`) shows how much shorter epochs would recover: 95% of recordings keep at least one epoch at 60 s and 88% at 90 s, against 59% at 180 s. Other possible fixes are 50%-overlapping windows (as in the preprint), computing DFA on concatenated clean data, or placing epochs between bad spans instead of on a fixed grid.

Results (Fig. 3).

- **MED vs IMW in meditators:** no DFA difference in any band ($|\Delta| < 0.02$, $p > .48$).
- **Meditators vs controls:** no DFA difference during IMW or MED in any band from theta to broadband ($p > .26$). During MED the CIs rule out differences larger than about 0.07 DFA units, roughly 10% of the mean exponent of ≈0.7. During IMW, where fewer recordings survive, they only rule out differences larger than about 0.15.
- **`med2` vs `med1breath` in meditators:** beta DFA is higher in the second block, by +0.052 to +0.059 ($p = .013$–$.034$ depending on pipeline). Broadband shows the same direction (+0.023 to +0.035, $p = .04$–$.20$) and alpha is not significant.
- **fEI:** no meditator–control differences. Within meditators, alpha fEI is slightly lower during MED than IMW (-0.04, $p = .036$–$.049$), and beta fEI likewise under the capped ICA only ($p = .045$). In controls, fEI falls from `med1breath` to `med2` in alpha and beta ($p = .013$–$.053$).
- **Gamma:** meditators' DFA and fEI tend to be higher than controls' during MED (+0.07 to +0.08, $p = .09$–$.11$). Given the ICA sensitivity of gamma power above, any such difference should be treated as artefact-prone until shown otherwise.

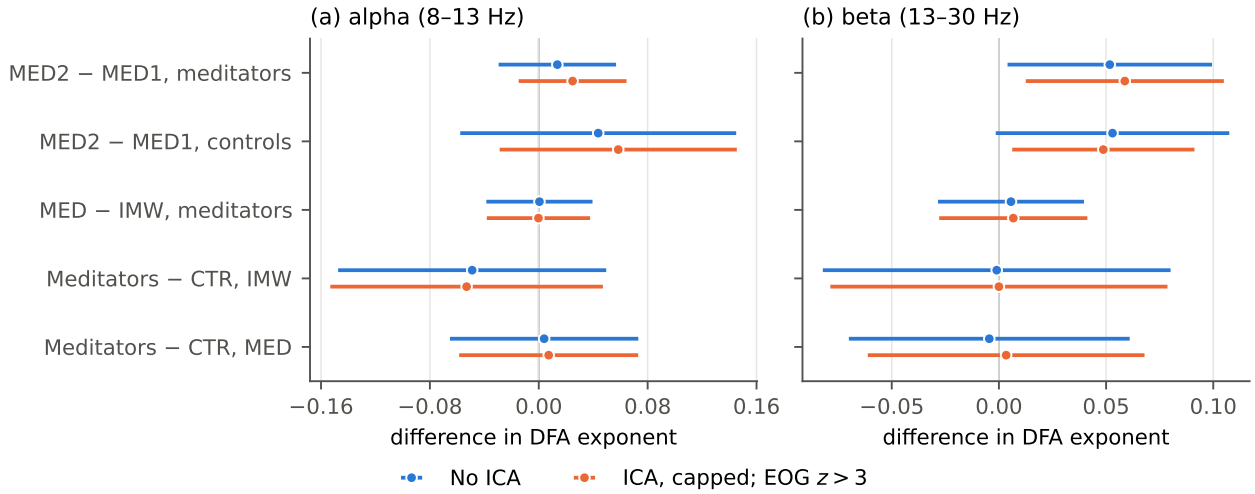


Figure 3: DFA contrasts (mixed-model estimates, 95% CIs) under both pipelines. "Meditators" means HYT, ISY and VIP pooled. MED − IMW pools both blocks of each condition.

The `med2` effect is probably time-on-task. `med2` always directly follows `med1breath`, and controls do breath focus in both blocks. If the rise reflected deepening practice, controls should not show it. They do, by the same amount:

- Beta DFA rises by +0.049 to +0.053 in controls ($p = .025$–$.057$; 14 controls, 6 with both blocks).
- The block × meditator interaction is essentially zero for beta (+0.009 and −0.004, $p > .8$), and negative but not significant for alpha.

The simplest account is that LRTCs in the beta envelope increase over the course of a 20-minute eyes-closed session. Drowsiness is the obvious candidate mechanism.

Multiplicity reinforces this reading. 120 contrasts were tested (6 tests × 5 bands × 2 metrics × 2 pipelines), and 10 reach $p < .05$, against about 6 expected by chance. They are scattered across bands and tests, and none would survive a Bonferroni or BH correction.

6 Comparison with "Near-critical brain dynamics track effortlessness during meditation"

Lewis-Healey, Kringelbach, Canales-Johnson and Laukkonen (bioRxiv, Aug 2026) recorded 128-channel EEG from 33 TM practitioners and 33 matched controls. Sessions consisted of a 10-minute rest, then 30 minutes of TM or silent counting, then a further 10-minute rest. Effort was rated retrospectively over time with Temporal Experience Tracing. Their criticality settings are essentially those used here: `crosci`, the same five bands, 3-minute windows, DFA up to 30 s, fEI with 5 s windows at 80% overlap, and fEI only where DFA > 0.6. They differ in a few respects:

- 50% window overlap.
- Band-specific minimum DFA windows (alpha 3.98 s, beta 2.24 s).
- Average reference, 0.5–45 Hz FIR filtering, FastICA with manual component rejection.
- Rejection on 5 s epochs.

The dataset is different. The TM group in `ds003969` has only 4 participants and is not in the paper's matched sample.

Agreement. The preprint finds *no robust between-group differences* in LRTCs or fEI, during either rest or practice. The replication agrees across all bands and both pipelines, with three meditation traditions instead of one. The preprint also finds no fEI condition effects after FDR correction; the few uncorrected fEI effects here would not survive such a correction either.

Apparent disagreement. Within meditators, the preprint finds TM increases alpha and beta DFA relative to rest both *before and after*. Here, MED does not differ from IMW. The comparisons are not equivalent:

- IMW is an active, effortful recall task rather than rest, so it is a weaker contrast for a hypothesis about "letting go".

- The preprint's pre/post bracketing controls for slow drift. The MED-vs-IMW comparison here does not: the sessions are counterbalanced across subjects but separated by about an hour, and much data has been lost.

The bracketing design matters. The `med2-vs-med1breath` analysis above shows that drift within a session is large enough (about +0.05 DFA units in beta over 10 minutes, in controls as well as meditators) to produce a spurious "deepening" effect when blocks are not counterbalanced.

What this data can and cannot say about effortlessness. The preprint's main finding is within-person: beta LRTCs rise as reported effort falls, more so in meditators. The strongest effect is beta (standardised $\beta = -0.28$), and meditators show higher beta LRTCs only in their lowest-effort windows. `ds003969` has no time-resolved phenomenology, so this cannot be tested directly. The nearest proxy, the move from effortful breath focus to the tradition's own practice, does show a beta-DFA increase. It is not specific to meditators, however, so it cannot be attributed to effortlessness. Two points follow:

- Time-on-task drift is a plausible confound for any effort–LRTC association, because reported effort also tends to fall over a session. The preprint addresses this with time-control models, and its effects survive (alpha $\beta = -0.47$, beta $\beta = -0.50$).
- The size of the drift seen here ($\sim$0.05 in 10 minutes) is a useful benchmark for how strong that control must be.

7 Summary and next steps

- **The pipeline reproduces the paper.** On the paper's matched sample and without ICA, per-subject gamma matches the published values at $r = 0.97$.
- **Gamma (meditators > controls): replicates**, by about 4 dB during MED. Automated ICA shrinks it to about 2.5 dB, still significant when pooled. The shrinkage is concentrated in 8 recordings, mostly meditators. The paper's manual ICA left parieto-occipital gamma essentially untouched.
- **Alpha (VIP > all other groups): replicates** under both pipelines.
- **Age matters.** Controls' gamma tends to fall with age while meditators' rises, so effect size depends on the age range. The paper's experience correlation looks confounded with age.
- **Criticality: null between groups**, consistent with the near-criticality preprint. The one within-meditator effect is most parsimoniously explained by time-on-task.

Next steps, in priority order:

1. Hand-label the ICA components behind the 8 recordings that lose more than 3 dB of gamma under automated ICA, looking specifically for EMG and saccadic spike-potential components. This decides whether the removed power is artefact or signal.
2. Reduce data loss in DFA/fEI with shorter or overlapping windows, or epochs placed between bad spans, and rerun the criticality contrasts.
3. Replace the uncorrected tests with cluster permutation tests, or at least FDR correction across bands and contrasts.

A The paper's spectra

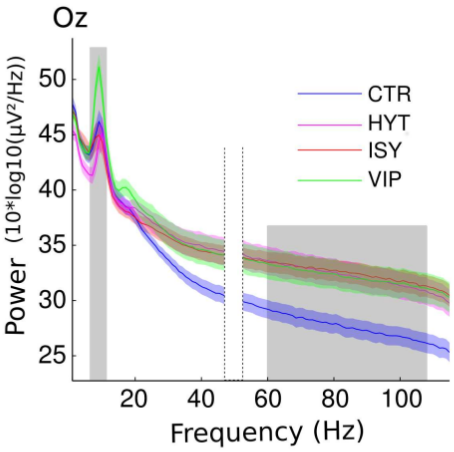


Fig 3. Spectral decomposition for occipital electrode Oz during the meditation condition in all 4 groups of subjects (CTR is the Control group; HYT is the Himalayan Yoga Tradition group; ISY is the Isha Shoonya Yoga group; VIP is the Vipassana group). Shaded regions indicate the standard error of the mean. The grey shaded area indicates the region of statistical differences between the groups after correction for multiple comparisons (see Material and Methods). The region around 50 Hz has been blanked out on the plot as it corresponds to line noise and has been excluded from analysis.

doi:10.1371/journal.pone.0170647.g003

Figure 4: Braboszcz et al. (2017), Fig. 3: Oz spectra by group (CC BY 4.0).

References

- Braboszcz C, Cahn BR, Levy J, Fernandez M, Delorme A (2017). Increased gamma brainwave amplitude compared to control in three different meditation traditions. *PLoS ONE* 12(1): e0170647. `https://pmc.ncbi.nlm.nih.gov/articles/PMC5261734/`
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- Dataset: OpenNeuro `ds003969` v1.0.0. `https://openneuro.org/datasets/ds003969/versions/1.0.0`